

Section 5

Systems Programming ECE3502

Sheet4, Q1) Pass 1

Location counter	Symbols	Instructions	reference
0000	SUM	Start	0000
0000		LDX	#0
0000+3=0003		LDA	#0
0003+3=0006		+LDB	#TABLE2
0006+4=000A	LOOP	BASE	TABLE2
000A		ADD	TABLE,X
000A+3=000D		ADD	TABLE2,X
000D+3=0010		TIX	COUNT
0010+3=0013	COUNT	JLT	LOOP
0013+3=0016		+STA	TOTAL
0016+4=001A		RSUB	
001A+3=001D		RESW	1
001D+(1*3) ¹⁰ =0020	TABLE	RESW	2000
0020+(2000*3) ¹⁰ =1790	TABLE2	RESW	2000
1790+(2000*3) ¹⁰ =2F00	TOTAL	RESW	1
2F00+(1*3) ¹⁰ =2F03		END	SUM

Symbol	Address
SUM	0000
LOOP	000A
COUNT	001D
TABLE	0020
TABLE2	1790
TOTAL	2F00

Sheet4, Q1) Pass 2

Location counter	Symbols	Inst.	Reference	Obj. code
0000	SUM	Start	0000	No Obj. code
0000		LDX	#0	050000
0003		LDA	#0	010000
0006		+LDB	#TABLE2	69101790
000A		BASE	TABLE2	No Obj. code
000A	LOOP	ADD	TABLE,X	
000D		ADD	TABLE2,X	
0010		TIX	COUNT	
0013		JLT	LOOP	
0016		+STA	TOTAL	
001A		RSUB		
001D	COUNT	RESW	1	
0020	TABLE	RESW	2000	
1790	TABLE2	RESW	2000	
2F00	TOTAL	RESW	1	
2F03		END	SUM	

LDX #0							
Opcode(6)	n	i	x	b	p	e	Disp(12)
04	0	1	0	0	0	0	0
0000 01	01		0000			0000 0000 0000	
050000							

LDA #0							
Opcode(6)	n	i	x	b	p	e	Disp(12)
00	0	1	0	0	0	0	0
0000 00	01		0000				0000 0000 0000
010000							

+LDB #TABLE2							
Opcode(6)	n	i	x	b	p	e	Address(20)
68	0	1	0	0	0	1	1790
0110 10	01	0001			0000 0001 0111 1001 0000		
69101790							

Sheet4, Q1) Pass 2.

Location counter	Symbols	Inst.	Reference	Obj. code
0000	SUM	Start	0000	No Obj. code
0000		LDX	#0	050000
0003		LDA	#0	010000
0006		+LDB	#TABLE2	69101790
000A		BASE	TABLE2	No Obj. code
000A	LOOP	ADD	TABLE,X	1BA013
000D		ADD	TABLE2,X	1BC000
0010		TIX	COUNT	2F200A
0013		JLT	LOOP	
0016		+STA	TOTAL	
001A		RSUB		
001D	COUNT	RESW	1	
0020		TABLE	RESW	2000
1790	TABLE2	RESW	2000	
2F00	TOTAL	RESW	1	
2F03		END	SUM	

ADD TABLE,X							
Opcode(6)	n	i	x	b	p	e	Disp(12)= TA – PC
18	1	1	1	0	1	0	0020-000D = 13
0001 10	1 1	1010				0000 0001 0011	
1BA013							

ADD TABLE2,X							
Opcode(6)	n	i	x	b	p	e	Disp(12) = = TA - PC B
18	1	1	1	1	0	0	1790-0010 = 1780 (6016) XX 1790-1790 = 0
0001 10	1	1	1100			0000 0000 0000	
1BC000							

TIX COUNT							
Opcode(6)	n	i	x	b	p	e	Disp(12)= TA – PC
2C	1	1	0	0	1	0	001D -0013 = A
0010 11	11	0010			0000 0000 1010		
2F200A							

Sheet4, Q1) Pass 2..

Location counter	Symbols	Inst.	Reference	Obj. code
0000	SUM	Start	0000	No Obj. code
0000		LDX	#0	050000
0003		LDA	#0	010000
0006		+LDB	#TABLE2	69101790
000A		BASE	TABLE2	No Obj. code
000A	LOOP	ADD	TABLE,X	1BA013
000D		ADD	TABLE2,X	1BC000
0010		TIX	COUNT	2F200A
0013		JLT	LOOP	3B2FF4
0016		+STA	TOTAL	0F102F00
001A		RSUB		4F0000
001D	COUNT	RESW	1	No Obj. code
0020	TABLE	RESW	2000	No Obj. code
1790	TABLE2	RESW	2000	No Obj. code
2F00	TOTAL	RESW	1	No Obj. code
2F03		END	SUM	No Obj. code

JLT LOOP							
Opcode(6)	n	i	x	b	p	e	Disp(12)= TA – PC
38	1	1	0	0	1	0	000A-0016= FF4 (-12)
0011 10	11	0010			1111 1111 0100		
3B2FF4							

+STA TOTAL							
Opcode(6)	n	i	x	b	p	e	Address(20)
0C	1	1	0	0	0	1	2F00
0000 11	11	0001			0000 0010 1111 0000 0000		
0F102F00							

RSUB							
Opcode(6)	n	i	x	b	p	e	Disp(12)
4C	1	1	0	0	0	0	0
0100 11	11	0000			0000 0000 0000		
4F0000							

H.T.M.E record

H. program name(6). Starting address(6). Length(6)

H.SUMXXX.000000.002F03

**T. Starting address of each t record(6) . Size of instruction(2).
obj. code of all inst(6)**

T.000000.1D.005000.001000.69101790.1BA013.1BC000.2F200A.3B2FF4.0F102F00.
4F0000

M. Location+1(6) . how many bit changed(2)

M.000007.05 (+LDB #TABLE2 69 1 01790)

M.000017.05 (+STA TOTAL 0F 1 02F00)

E. Starting address(6)

E.000000

Sheet4, Q2)

Translate (by hand) the following assembly program to SIC/XE object code. Which contains H record, T record, M record and E record.

00000	COPY	Start	0000
000		LDS	#3
0003		LDT	#300
0006		LDX	#0
0009		LDA	ALPHA,X
000C	ADDLP	ADD	BETA,X
000F		STA	GAMMA,X
0012		ADDR	S,X
0015		COMPR	X,T
0017		JLT	ADDLP
0019		RESW	100
001C	ALPHA		
0148	BETA		
0274	GAMMA		
03A0		END	COPY

Symbols	Address
ADDLP	000C
ALPHA	001C
BETA	0148
GAMMA	03A0

ADDR S,X		
Opcode(8)	R1(4)	R2(4)
18	4	1
0001 1000	0100	0001
1841		